📚 CLASS 10 SCIENCE: OUR ENVIRONMENT

Subject: Biology / Environment | **Language:** Hinglish (Easy & Clear) | **Format:** Topper's Revision Notes

1. What is an Ecosystem? (Definition)

English: All interacting organisms in an area together with the non-living components of the environment form an ecosystem.

Hinglish meaning: Ek aisa system jisme saare living organisms (plants, animals) aur non-living things (air, water, soil) ek saath mil kar interact karte hain, use **Ecosystem** kehte hain.

> 💡 **Real-life Example:** Ek pond (talab), forest, aquarium, ya crop field ek ecosystem hai. Aquarium ek **Artificial (Man-made) Ecosystem** hai, jabki forest ek **Natural Ecosystem** hai.

2. Components of an Ecosystem

Ecosystem ke do main parts hote hain:

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      ECOSYSTEM / \ Biotic Components Abiotic Components (Living Organisms) (Non-living Factors) - Producers - Temperature - Consumers - Rainfall - Decomposers - Soil, Air, Light
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A. Biotic Components (Living things)

- **Producers (Autotrophs):** Jo apna khana khud banate hain using sunlight (Photosynthesis).
  Example: Green plants, Blue-green algae.
- **Consumers (Heterotrophs):** Jo apne khana ke liye producers par depend hote hain.
  Types: Herbivores (Cow), Carnivores (Lion), Omnivores (Human), Parasites (Cuscuta).
- **Decomposers (Saprotrophs):** Dead plants aur animals ko break down karke simple substances mein convert karte hain.
  Example: Fungi, Bacteria. (Nature ke scavengers).

Components of an Ecosystem (Contd.)

B. Abiotic Components (Non-living things)

Physical factors jaise **Temperature, Light, Wind, Soil, Water, aur Minerals**. Yeh decide karte hain ki wahan kis tarah ke plants aur animals survive karenge.

3. Food Chain (Aahaar Shrinkhala)

Definition: A linear network of links in a food web starting from producer organisms and ending at apex predators.

Hinglish Concept: "Who eats whom" ki ek seedhi line ko food chain kehte hain jisme energy ek organism se dusre organism mein transfer hoti hai.

> 🌿 **Simple Food Chain Example:**
> Grass (Producer) → Deer (Herbivore/Primary Consumer) → Lion (Carnivore/Secondary Consumer)

Key Characteristics of Food Chain:

- Energy flow hamesha **Unidirectional** (ek hi direction mein) hoti hai.
- Yeh 3 to 4 steps (trophic levels) se zyada lambi nahi hoti kyunki aage jaate-jaate energy khatam ho jati hai.

4. Food Web (Aahaar Jaal)

Definition: An intricate network of many food chains intersecting with each other in an ecosystem.

Hinglish Concept: Jab bohot saari food chains aapas mein jud kar ek complex jaal (web) bana leti hain, toh use **Food Web** kehte hain. Yahan ek animal ke paas multiple khane ke options hote hain.

Why important? Agar kisi ek food chain mein koi animal mar jaye, toh ecosystem collapse nahi hota kyunki organisms ke paas alternate food options hote hain.

5. Trophic Levels (Poshak Star)

Food chain ke har step ya level ko **Trophic Level** kehte hain.

- **T_1 (First Trophic Level):** Producers (Plants)
- **T_2 (Second Trophic Level):** Herbivores (Primary Consumers)
- **T_3 (Third Trophic Level):** Carnivores (Secondary Consumers)
- **T_4 (Fourth Trophic Level):** Large Carnivores (Tertiary Consumers)

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     T4: Large Carnivores (Tertiary Consumers) ▲ T3: Carnivores (Secondary Consumers) ▲ T2: Herbivores (Primary Consumers) ▲ T1: Producers (Plants / Autotrophs)
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6. 10% Law of Energy Flow (Lindeman's Law)

Definition: Only 10% of the energy entering a trophic level is available for transfer to the next trophic level. Baki 90% energy organism apni life activities (respiration, growth) mein use kar leta hai aur heat ke roop mein lose ho jati hai.

> **Formula:** Energy at next level = 10% of Energy at previous level

10% Law: Solved Example & Trick

Agar Producer (T_1) ke paas **10,000 J** energy hai:

- **T_1 (Plants)** = 10,000 J
- **T_2 (Deer)** = 10% of 10,000 = 1,000 J
- **T_3 (Lion)** = 10% of 1,000 = 100 J
- **T_4** = 10 J

🚀 **Exam Shortcut Trick:** Har next level par jaane par ek zero (0) hata do!
10,000 → 1,000 → 100 → 10.

7. Biological Magnification (Biomagnification)

Definition: The progressive accumulation of harmful, non-biodegradable chemicals (like pesticides, heavy metals) at each succeeding trophic level in a food chain.

Hinglish Concept: Jaise-jaise hum food chain mein upar ki taraf (Producers → Consumers) jaate hain, harmful chemicals ki concentration barhti jati hai.

⚠️ **Exam Special Note:** Sabse zyada toxic chemicals **Top Carnivores (jaise Human ya Lion)** ke andar accumulate hote hain!

Biomagnification: Real-Life Application

Real-life Example:

- Fasal par DDT (pesticide) chhidka gaya.
- Plants ne soil/water se absorb kiya (Low concentration).
- Herbivores ne khaya (Concentration badhi).
- Carnivores/Humans ne khaya (Maximum concentration/Toxic level).

8. Ozone Layer Depletion

A. What is Ozone?

- **Formula:** O_3 (An isotope of oxygen).
- **Location:** Stratosphere (atmosphere ki ek layer jo zameen se lagbhag 15-50 km upar hai).
- **Role:** Yeh sun se aane wali harmful **Ultraviolet (UV) Radiations** ko absorb karti hai. UV rays se skin cancer, cataract, aur immune system weak ho sakta hai.

Ozone Layer Depletion (Contd.)

B. Ozone Hole & Causes

Hinglish Concept: Ozone layer patli ho rahi hai (jise ozone hole kehte hain) jiska main reason hai **CFCs (Chlorofluorocarbons)**.

Source of CFCs: Refrigerators, ACs, aur aerosol sprays.

Mechanism: CFCs jab stratosphere mein pahunchte hain, toh chlorine atoms release karte hain jo ozone (O_3) ko tod kar oxygen (O_2) mein badal dete hain.

> 💡 **UNEP Agreement:** 1987 mein **UNEP (United Nations Environment Programme)** ne ye decide kiya ki duniya bhar ke countries CFC production ko freeze karein under **Montreal Protocol**.

9. Waste Management & Disposal

Hum jo waste produce karte hain, use do main categories mein divide kiya gaya hai:

Feature	Biodegradable Waste	Non-Biodegradable Waste
Meaning	Wastes broken down by biological processes (microorganisms).	Wastes that **cannot** be broken down easily.
Time taken	Fast (few days to weeks).	Extremely slow (thousands of years).
Impact	Eco-friendly if disposed; bad smell if accumulated.	Causes pollution, biomagnification, harms soil.
Examples	Vegetable peels, paper, leftover food, cow dung.	Plastics, polythene bags, glass, synthetic fibers.

Waste Management Strategies

Wastes ko manage karne ke liye **3 R's principle** follow kiya jata hai:

- **Reduce:** Kam se kam waste generate karna (jaise paani ya bijli bachana, extra packaging avoid karna).
- **Reuse:** Cheezon ko dobara use karna bina unhe fenke (jaise plastic bottles ko plant pot ki tarah use karna).
- **Recycle:** Waste material ko process karke nayi useful cheezein banana (jaise old paper se recycled paper).

> **Note:** "Reduce" sabse best option hai kyunki isme energy aur resources ki sabse kam wastage hoti hai.

10. Diagrams & Flowcharts Explanation

Text-Based Diagram: Energy Flow in an Ecosystem

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      Sun ☀️ | ▼ (Light Energy) [ Producers (Plants) ] ──(90% Heat Loss)──> | ▼ (10% Energy Passed) [ Herbivores (Deer) ] ──(90% Heat Loss)──> | ▼ (10% Energy Passed) [ Carnivores (Tiger) ] ──(90% Heat Loss)──>
```

Explanation: Energy ka flow **ek-tarfa (unidirectional)** hota hai. Sun se energy plants mein aati hai, aur phir step-by-step consumer levels par kam hoti jati hai. Energy kabhi wapas Sun par nahi jaati.

11. Exam Short Tricks & Mnemonics

🚀 **Topper's Quick Tricks:**

- **Ozone Formula Trick:** O ke baad 3 aata hai → O_3.
- **10% Law Trick:** Har level par ek zero (0) hata do! (10,000 → 1,000 → 100 → 10).
- **Biomagnification Tip:** Yaad rakho **"Top = Toxic"**. Top trophic level par toxic chemical sabse zyada hoga.
- **Biodegradable vs Non-bio:** Bio = Living (jo decompose ho jaye). Non-bio = Man-made plastics (jo kabhi khatam na ho).

Frequently Asked Questions (Board Exam Focus)

- **Q1. Why are food chains generally limited to 3 or 4 trophic levels?**
  Ans: Because according to the 10% law, very little energy is available for the next level after 3-4 transfers, making survival impossible beyond that.
- **Q2. What would happen if we kill all organisms in one trophic level?**
  Ans: The preceding level population will grow uncontrollably, and the succeeding level will starve and die, collapsing the ecosystem.
- **Q3. Why is plastic non-biodegradable?**
  Ans: Because microorganisms lack the specific enzymes required to break down synthetic polymers found in plastics.

⚡ Quick Revision Checklist (Last-Minute Summary)

- [ ] **Ecosystem:** Biotic (living) + Abiotic (non-living).
- [ ] **Producers:** Autotrophs (Plants).
- [ ] **Decomposers:** Cleaners of nature (Bacteria/Fungi).
- [ ] **10% Law:** Lindeman ne diya tha; sirf 10% energy next level par jati hai.
- [ ] **Biomagnification:** Chemicals ki concentration trophic level ke sath barhti hai.
- [ ] **Ozone (O_3):** UV rays se bachata hai; CFCs isko destroy karte hain.
- [ ] **Montreal Protocol:** Related to Ozone layer protection.
- [ ] **3 R's:** Reduce, Reuse, Recycle.

🎉 All the Best for Your Board Exams! Revise well!